

Predicting nitrification status in aerobic membrane bioreactors by interpretable machine learning models: Proof of concept for advancing greywater reuse through process monitoring with cross-scenario model validation

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ABSTRACT

Membrane bioreactor (MBR) systems for on-site greywater treatment and reuse have been widely investigated, where robust nitrification monitoring is critical to prevent operational disruptions and maintain effluent standards. This research proposed an interpretable machine learning-based framework to predict nitrification efficacy (classified as "sufficient" or "insufficient") in MBR systems, involving model-based and post hoc interpretability. Three interpretable machine learning algorithms—logistic regression (LR), random forest (RF), and extreme gradient boosting (XGB)—were systematically evaluated, accounting for model bias-variance trade-offs during binary classification. Six input features were selected based on direct measurability, compatibility with commercial sensors, and operational relevance. Model optimization criteria focused on maximizing precision to prevent erroneous controls for insufficient nitrification, achieved by low false positive rate (FPR: misclassified insufficient nitrification) and high true positive rate (TPR: correctly identified sufficient nitrification). Results showed that both LR and XGB achieved identical precision scores > 0.85 under standard conditions. Cross-scenario validation demonstrated RF's superior generalizability (precision=0.87) when predicting nitrification in biocarrier-amended MBRs using training data from non-biocarrier systems, reflecting the potential transferability of the model. The post hoc interpretability and stability analysis further identified prediction sensitivity to data limitations and distributional biases, providing critical insights for improving and even properly deploying models in practice.

1. Introduction

Greywater, defined as domestic wastewater excluding toilet contributions, represents around 70 % of residential sewage, with daily yields ranging from 20 to 220 L per capita (Shaikh and Ahammed, 2020). Its on-site reuse offers a critical strategy for water conservation, urban water resilience improvement, and sustainable urban development (Liu et al., 2023; Maggiotto, 2022). Membrane

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2. Vật liệu và phương pháp

2.1. Phương pháp thực nghiệm

Dự đoán trạng thái nitrat hóa trong bể phản ứng sinh học màng hiếu khí bằng các mô hình học máy có thể giải thích

1. Giới thiệu

Fig. 1. Thiết lập thực nghiệm và các giai đoạn vận hành của MBR.

2.2. Mô hình hóa dựa trên dữ liệu

bioreactors (MBRs) have emerged as a leading technology for such decentralized treatment, owing to their compact footprint and consistent production of high-quality effluent suitable for reuse (Gao et al., 2025). MBR systems synergistically combine biological processes—including heterotrophic oxidation and autotrophic nitrification—with membrane filtration for effective solids separation (Boleydei and Vaneeckhaute, 2024).

Nitrification—converting ammonium ($\text{NH}_4^+\text{-N}$) into nitrite ($\text{NO}_2^-\text{-N}$) and then nitrate ($\text{NO}_3^-\text{-N}$)—is critical in MBRs to meet water reuse standards. Centralized plants typically use the proportional integral differential (PID) control strategy to maintain fixed dissolved oxygen (DO) setpoints, yet this strategy adapts poorly to influent fluctuations and incurs high energy costs (Gu et al., 2023). On-site greywater systems face amplified load variability: daily/seasonal usage patterns cause sustained low loads punctuated by sharp peaks, while highly variable water quality challenges empirical control parameters (DelaPaz-Ruíz et al., 2024; Hassan et al., 2024). Although PID adjusts aeration valves, its static DO targets lack real-time responsiveness to dynamic $\text{NH}_4^+\text{-N}$ loads (e.g., dynamic adjustment based on the $\text{NH}_4^+\text{-N}$ load) (Li et al., 2022; Shi et al., 2024). Furthermore, PID relies on submerged DO probes, which are prone to fouling and drift in small-scale MBRs due to continuous aeration and sludge accumulation. This compromises measurement reliability, potentially masking sensor failure as load variation.

Reliable nitrification monitoring remains challenging for decentralized MBRs, where inconsistent efficiency risks process stability despite compliant effluent (observed in the current trial). Notably, real-time monitoring of nitrification processes within MBRs—particularly under decentralized operational scenarios—remains unaddressed in existing studies. Commercial $\text{NH}_4^+\text{-N}/\text{NO}_2^-\text{-N}/\text{NO}_3^-\text{-N}$ sensors are often impractical for installing in the reactor: high sludge concentrations cause rapid fouling and maintenance burdens, increasing operational costs (Huang et al., 2024). Organic matter, suspended solids, and ionic interference further distort readings, while sensor response lags fail to capture rapid nitrogen dynamics—hindering real-time control.

Machine learning (ML) offers promising data-driven solutions for monitoring MBR nitrification, using effluent water quality parameters to assess performance without intrusive sensors, avoiding sensor installation in the reactors with challenges mentioned above (Van de Walle et al., 2023). These methods enable rapid, low-cost, and real-time monitoring—facilitating prompt process interventions and enhancing safety (Bahramian et al., 2023). Compared to mechanistic models, ML requires simpler inputs and incurs minimal maintenance, ideal for on-site systems (Duarte et al., 2024). Given greywater treatment’s critical need for regulatory and public trust, prioritizing interpretable ML methods provides transparent prediction rationales for operator action and model refinement (Allen et al., 2024). Additionally, utilizing interpretable ML methods allows us to further understand the limitations of current training

Fig. 2 trình bày lưu đồ toàn diện được áp dụng để phát triển các mô hình dựa trên dữ liệu nhằm giám sát quá trình nitrat hóa trong MBR (bể phản ứng sinh học màng). Ngoài ra, phân tích kết quả và định hướng tương lai được trình bày chi tiết trong Mục 3 và 4.

2.2.1. Phân loại nhị phân: đánh giá quá trình nitrat hóa

Để dự đoán hiệu quả trạng thái nitrat hóa, mối quan hệ giữa nồng độ $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$ và $\text{NH}_4^+\text{-N}$ trong bể phản ứng cần phải được hiểu rõ. Trong quá trình nitrat hóa và khử nitrat đồng thời bên trong MBR (bể phản ứng sinh học màng), nồng độ $\text{NO}_3^-\text{-N}$ thường vượt quá nồng độ $\text{NO}_2^-\text{-N}$ và $\text{NH}_4^+\text{-N}$ trong điều kiện sục khí đầy đủ, vốn là yếu tố then chốt cho quá trình chuyển hóa và loại bỏ nitơ hiệu quả (Huang et al., 2022; Paetkau and Cicek, 2011). Do đó, nitrat hóa đầy đủ được định nghĩa là điều kiện mà trong đó nồng độ $\text{NO}_3^-\text{-N}$ vượt quá tổng nồng độ của $\text{NO}_2^-\text{-N}$ và $\text{NH}_4^+\text{-N}$, được gán nhãn là “Positive”; ngược lại, điều kiện này được gán nhãn là “Negative”. Mục tiêu chính của mô hình dựa trên dữ liệu nhằm dự đoán quy trình xử lý là phân loại xem quá trình nitrat hóa là đầy đủ hay không đầy đủ.

2.2.2. Phân tích định tính việc lựa chọn biến đầu vào

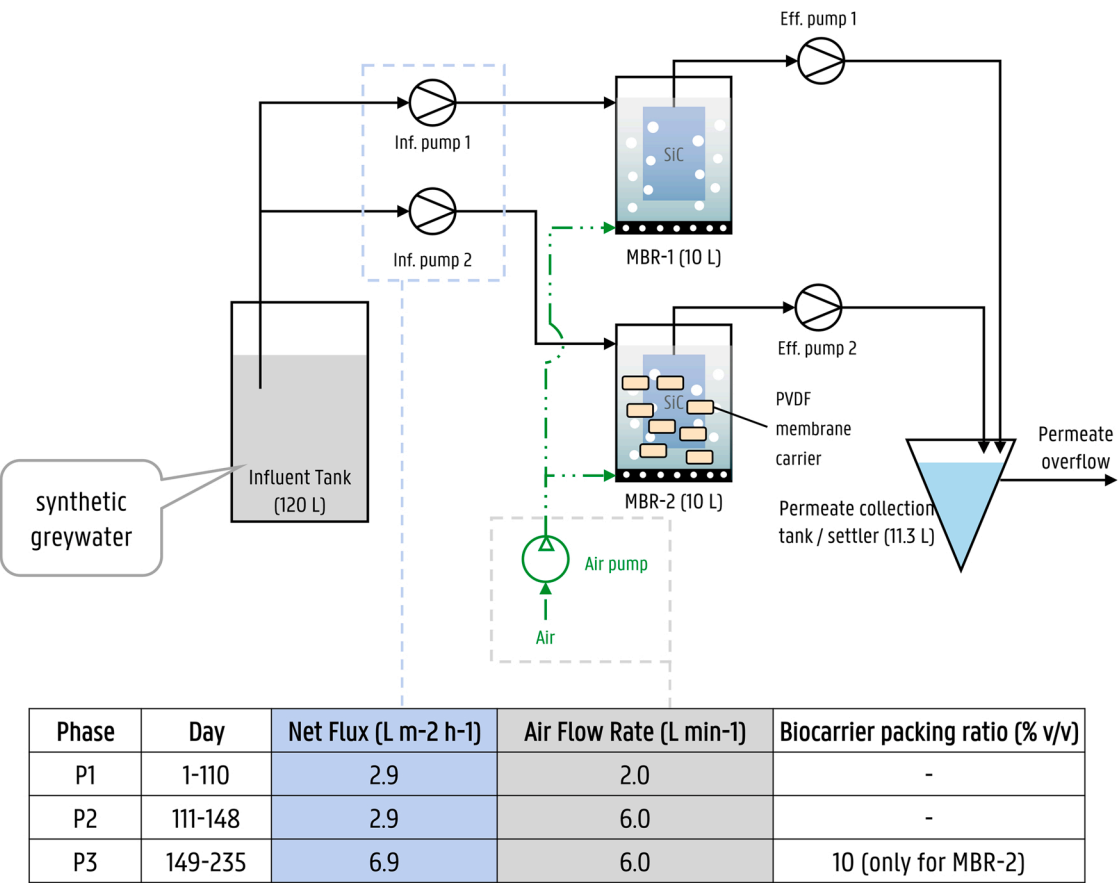


Fig. 1. Experimental setup and operation phases of MBR.

strategies and provide guidance for model improvement.

Previous studies on data-driven prediction in MBR-based wastewater treatment have primarily focused on effluent quality, targeted pollutant removal efficiency, and membrane fouling (Muniz de Queiroz et al., 2025; Shi et al., 2022; Zhong et al., 2022). However, relying solely on effluent monitoring provides limited process controllability and fails to enable timely interventions during abnormal operational events. This approach is also labor-intensive, time-consuming, and inadequate for tracing underlying process flaws before effluent quality violations occur (Moretti et al., 2024). In contrast, real-time process monitoring enables proactive control, enhances system resilience, and supports early fault detection—critical for ensuring consistent water quality across diverse reuse applications.

This work aims to develop an interpretable ML model to monitor the nitrification process in MBRs via effluent parameters, reducing the difficulty of input feature acquisition in practice. Features were selected based on direct measurability, sensor compatibility, and operational relevance. To mitigate overfitting and balance bias-variance tradeoffs, three interpretable algorithms—logistic regression (LR), random forest (RF), and extreme gradient boosting (XGB)—were evaluated, simultaneously considering the bias and variance of each algorithm. Model training prioritized operational stability by maximizing precision, depending on the higher true positive rate (TPR) and lower false positive rate (FPR), trained on data from a suspended-sludge MBR. Furthermore, SHapley Additive exPlanations (SHAP) analysis, feature importance, and pairwise assessments elucidated factor-prediction linkages and informed model refinement. The models were further validated on a hybrid system, integrating suspended sludge and polyvinylidene fluoride (PVDF) biocarriers, performing simultaneously nitrification-denitrification, demonstrating cross-scenario prediction applicability. Finally, an automated aeration control strategy is proposed to operationalize the monitoring framework.

2. Material and methods

2.1. Experiment method

Synthetic greywater was prepared following Ongena et al. (2023) using Korean commercial personal care products, detergents, and

2.2.6. Khả năng diễn giải dựa trên mô hình: các thuật toán ML (học máy) có khả năng diễn giải

2.2.3. Tiền xử lý dữ liệu

2.2.4. Chiến lược phân chia tập dữ liệu

2.2.5. Phân tích tương quan thống kê để xác định các biến đầu vào

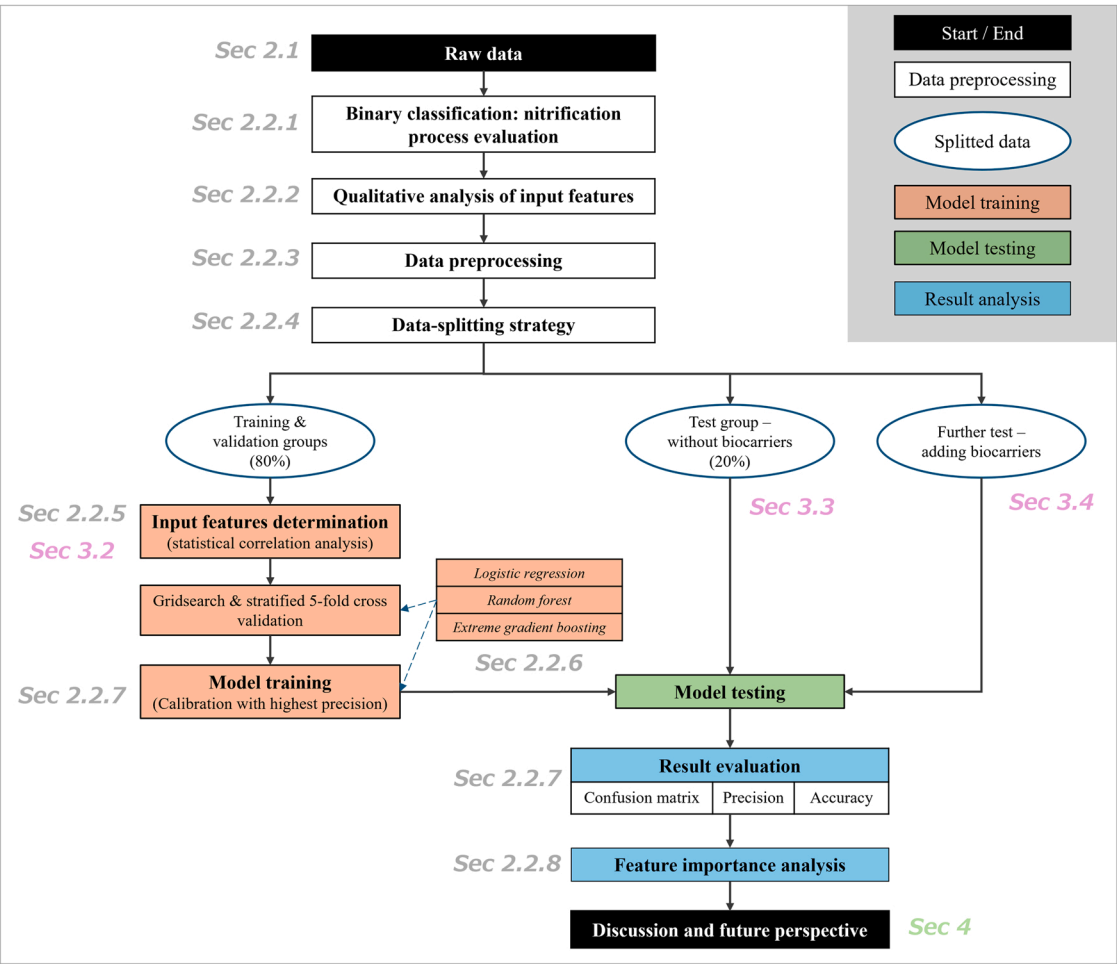


Fig. 2. Flowchart of data-driven model development and evaluation for nitrification process monitoring.

literature-derived compositions. The overall synthetic greywater contained $405 \pm 70 \text{ mg L}^{-1}$ of chemical oxygen demand (COD), $20 \pm 3 \text{ mg L}^{-1}$ of $\text{NH}_4^+ - \text{N}$ and $21 \pm 5 \text{ mg L}^{-1}$ of total nitrogen (TN). The remainder of the COD and TN not provided by the personal care products and detergents was approximated by adding acetate and ammonium chloride. Fresh batches were prepared daily with tap water.

Two parallel 10 L MBRs were inoculated with activated sludge sampled from an Anaerobic-Anoxic-Oxic wastewater treatment plant (Incheon, Korea) and operated for 235 days. Each contained two 0.56- μm SiC flat-sheet membranes (0.165 m² surface area; Fig. 1). Continuous aeration maintained DO at $5.1 \pm 2.2 \text{ mg L}^{-1}$. Greywater feeding and permeate extraction were conducted continuously, with timer-controlled relaxation (0.06 L min⁻¹). The operational strategy for the two parallel MBRs involved three phases of adjusted net flux and aeration to enhance greywater treatment, particularly ammonia removal (Fig. 1). Initially low ammonia removal under a flux of 2.9 L/(m²·h) and aeration of 2.0 L/min prompted an increase in aeration to 6.0 L/min in Phase 2 to improve nitrification efficiency. Phase 3 involved an increase in flux to 6.9 L/(m²·h) to further test hydraulic stability. Biocarriers (10 % packing ratio) were added to MBR-2 in phase 3 to try to reduce membrane fouling under high flux and promote anoxic zones for enhancing simultaneous nitrification-denitrification, enabling comparative performance evaluation with MBR-1. The 10 % packing ratio was selected to balance fouling control with operational safety, avoiding abrasion and energy issues associated with higher loads (Noor et al., 2023; Rahman et al., 2023). These intentional variations in flux, aeration, and biocarriers addition diversified operational conditions, enriching the dataset to enhance the machine learning model’s generalizability across multiple scenarios.

Influent, effluent and reactor water quality were analyzed daily as described in [supplementary information](#) (SI) 1.

2.2. Data-driven modelling

Fig. 2 presents the comprehensive flow-chart adopted for developing data-driven models to monitor the nitrification process in the MBR. Additionally, result analysis and future perspectives are elaborated in [Sections 3 and 4](#).

2.2.1. Binary classification: nitrification process evaluation

To effectively predict nitrification status, the relationship between the concentrations of $\text{NO}_3^- - \text{N}$, $\text{NO}_2^- - \text{N}$, and $\text{NH}_4^+ - \text{N}$ in the reactor must be understood. In the simultaneous nitrification and denitrification process within the MBR, the concentration of $\text{NO}_3^- - \text{N}$ typically exceeds that of $\text{NO}_2^- - \text{N}$ and $\text{NH}_4^+ - \text{N}$ under sufficient aeration, which is a key factor for effective nitrogen conversion and removal (Huang et al., 2022; Paetkau and Cicek, 2011). Therefore, sufficient nitrification was defined as the condition in which the concentration of $\text{NO}_3^- - \text{N}$ exceeds the combined concentrations of $\text{NO}_2^- - \text{N}$ and $\text{NH}_4^+ - \text{N}$, labeled as “Positive”; otherwise, the condition was labeled as “Negative”. The primary goal of the data-driven model for treatment process prediction was to classify whether nitrification was sufficient or insufficient.

2.2.2. Qualitative analysis of inputs selection

Three data categories were collected during the experiment: 1 > Hydraulic: hydraulic retention time (HRT) and sludge retention time (SRT); 2 > Operational: transmembrane pressure (TMP), air/influent/effluent flow rates, gross/net flux, flux recovery; 3 > Water quality: COD, TN, $\text{NH}_4^+ - \text{N}$, $\text{NO}_3^- - \text{N}$, $\text{NO}_2^- - \text{N}$ (in influent, effluent, and mixed liquor in the reactor), TSS (influent/effluent), and DO, MLSS, MLVSS in the reactor. Removal efficiency was also calculated for each parameter. Input features were selected from the perspective of practical deployment, including: 1> Direct measurability, 2> Commercial sensor compatibility, 3> Operational relevance (e.g., TMP).

Indirect variables (SRT, removal efficiencies) were excluded. Water quality parameters (DO, COD, TN, $\text{NH}_4^+ - \text{N}$, $\text{NO}_3^- - \text{N}$, $\text{NO}_2^- - \text{N}$) and operational factors (flow rates, air flow, TMP) were initially considered. However, > 50 % DO data absence led to its exclusion—simulating probe failure scenarios. Air flow rate was retained as an oxygen proxy. Online sensor compatibility was prioritized to enhance model applicability and robustness. Influent water quality was excluded due to sensor reliability concerns: complex matrices cause fouling on sensors and sensor degradation (e.g., Ag/AgCl electrodes dissolution) (Ching et al., 2022; Haimi et al., 2013). Effluent $\text{NO}_2^- - \text{N}$ was also excluded due to measurement instability versus stable $\text{NO}_3^- - \text{N}$. TMP was included to monitor membrane performance, contributing to long-term operation monitoring. Finally, the dataset was refined by removing inappropriate variables and simplifying the inputs to facilitate dataset splitting.

2.2.3. Data preprocessing

From 128 datasets collected over 235 days from dual MBRs, eight incomplete entries were excluded. Biocarriers were added to one reactor in Phase 3 to enable denitrification (Fig. 1), altering parameter relationships. To evaluate cross-scenario applicability, 23 data groups from Phase 3 were used as an independent test group for models trained exclusively on 97 no-biocarrier data groups (activated sludge only). The 97 data groups were utilized for training, validation, and testing of the model’s performance in predicting nitrification with activated sludge only. This approach assessed performance under altered operational conditions. Given inherent influent variability in onsite greywater systems, outliers were intentionally retained to evaluate model robustness against operational fluctuations, reflecting real-world dynamics.

2.2.4. Dataset splitting strategy

The preprocessed dataset comprised 120 groups of data, daily measurements reflecting operational manipulations. The data exhibited an inherent imbalance in label distribution due to controlled changes in treatment conditions. For each reactor, the airflow

Đề dự đoán trạng thái nitrat hóa trong MBR và đảm bảo vận hành ổn định, nghiên cứu này không chỉ xem xét độ chính xác thống kê của các phương pháp ML (học máy) mà còn tích hợp các đặc tính riêng của từng thuật toán. Trong ML (học máy), độ chệch bất nguồn từ các giả định đơn giản hóa, trong khi phương sai phản ánh độ nhạy đối với các biến động trong dữ liệu huấn luyện (Reynaert et al., 2023). Việc giảm thiểu một yếu tố thường làm gia tăng yếu tố còn lại (Belkin et al., 2019).

Fig. 3. Chiến lược phân chia cho các nhóm huấn luyện, thẩm định và kiểm tra.

2.2.7. Các chỉ số đánh giá và tiêu chí tinh chỉnh siêu tham số

Việc giảm thiểu FPR đóng vai trò rất quan trọng nhằm đảm bảo chất lượng nước đầu ra an toàn. Ví dụ, trong một hệ thống điều khiển tích hợp mô hình giám sát quá trình nitrat hóa này, khi quá trình nitrat hóa không đầy đủ xảy ra, cần phải cấp thêm không khí. Tuy nhiên, nếu xảy ra FP, mô hình dự đoán sai là nitrat hóa đầy đủ, dẫn đến việc hệ thống điều khiển không bổ sung thêm oxy và tạo ra nước đầu ra không an toàn, không phù hợp để tái sử dụng. Trong khi đó, TPR cao hơn là cần thiết để phát hiện quá trình nitrat hóa đầy đủ trong điều kiện vận hành thông thường, giúp giảm chi phí vận hành bằng cách ngăn chặn việc cấp oxy không cần thiết do FN gây ra.

Bên cạnh đó, việc chỉ tối ưu hóa để có FPR thấp có thể làm giảm mạnh TPR, như đã được chỉ ra trong các nghiên cứu trước đây (Reynaert et al., 2023; Salvia et al., 2021; Yang et al., 2019). Do đó, một phương pháp tiếp cận cân bằng có tính đến cả TPR (Eq. 1) và FPR (Eq. 2) đã được áp dụng, với độ chuẩn xác (Precision) (Eq. 3) đóng vai trò là chỉ số tối ưu hóa then chốt. Độ chuẩn xác phản ánh tỷ lệ các dự đoán TP trong tổng số tất cả các dự đoán dương tính. TPR, FPR và độ chuẩn xác được tính toán theo các phương trình sau:

Cuối cùng, tính ổn định của hiệu năng mô hình được đánh giá trên tập dữ liệu thử nghiệm bằng cách tính khoảng tin cậy 95 % cho độ chính xác, độ chuẩn xác, độ thu hồi (Recall) và điểm F1 bằng phương pháp tái lấy mẫu bootstrap 1000 lần, tuân theo các thực hành chuẩn trong kiểm chứng học máy.

Mặc dù các phương pháp dựa trên cây quyết định có khả năng giải thích vốn có, các mô hình phức tạp vẫn đòi hỏi việc giải thích hậu nghiệm (post hoc) (Murdoch et al., 2019).

rate was maintained at 2.0 L/min until day 111 and increased to 6.0 L/min thereafter to enhance the nitrification treatment. Similarly, the influent flow rate was predominantly 11.7 ML/min before the 149th day and increased to 20.6 ML/min afterward. Given these temporal shifts, a simple chronological split of the data could not fairly evaluate model performance. To ensure a representative assessment, the training and test sets were constructed strategically to encompass these operational variations, as illustrated in Fig. 3.

Given the imbalanced air flow and influent flow rates, every fourth data point was selected to form the test set, constituting 20 % of the data (8 positive & 11 negative labels). The remaining 80 % (28 positive & 50 negative labels) were used for training and validation, maintaining distinct distributional characteristics to assess model generalization. Hyperparameter optimization employed stratified 5-fold cross-validation (Scikit-Learn's *StratifiedKFold*), ensuring balanced class representation in each fold for enhanced reliability with imbalanced dataset (Szeghalmy and Fazekas, 2023; Zeng and Martinez, 2000). Additionally, 23 datasets from the biocarrier-added condition (Section 2.2 ; 15 positive and 8 negative labels) were used to test cross-scenario robustness under divergent label distributions. Section 2.1

2.2.5. Statistical correlation analysis for inputs determination

Based on the preliminary input feature selections by qualitative analysis in Section 2.2.2, Spearman's rank correlation was used to assess statistical relationships between inputs and outputs. This nonparametric method quantifies the strength of monotonic associations between variables without assuming normal distribution, making it suitable for nonlinear yet monotonic classification contexts (Hastie et al., 2009). Statistical significance was determined using p-values. Specifically, the p-value indicates the probability of obtaining the observed Spearman correlation coefficient, or a more extreme value, under the null hypothesis that no monotonic relationship exists between the variables. A p-value < 0.05 was used to reject the null hypothesis and confirm a significant correlation. The "spearmanr" function (SciPy, Python) computed coefficients and p-values for all variable pairs. (Van Rossum and Drake, 2009).

2.2.6. Model-based interpretability: interpretable ML algorithms

To predict the nitrification status in the MBR and ensure stable operations, this study considered not only the statistical accuracy of ML methods but also integrated algorithm-specific characteristics. In ML, bias stems from simplifying assumptions, while variance reflects sensitivity to fluctuations in the training data (Reynaert et al., 2023). Minimizing one typically increases the other (Belkin et al., 2019).

Phân tích SHAP được triển khai nhằm tận dụng lý thuyết trò chơi để gán cho mỗi đặc trưng một giá trị quan trọng đối với các dự đoán cụ thể. SHAP

Ngoài ra, mối quan hệ giữa các cặp đặc trưng được trực quan hóa thông qua ước lượng mật độ nhân (KDE) và biểu đồ đặc trưng theo cặp. KDE có thể xấp xỉ phi tham số các hàm mật độ xác suất liên tục bằng cách làm mịn các điểm dữ liệu quan sát được thông qua các hàm nhân đối xứng và các tham số băng thông (Scott, 2015). Biểu đồ đặc trưng theo cặp, được triển khai dưới dạng các ma trận biểu đồ phân tán, thể hiện các dạng tương quan và phân cụm giữa các đặc trưng (Pekalska et al., 2005). Phương pháp này có thể giúp hiểu rõ hiệu năng phân loại của mô hình bằng cách trực quan hóa ma trận nhầm lẫn (Waskom, 2021).

3. Phân tích kết quả: dự đoán mô hình dựa trên dữ liệu

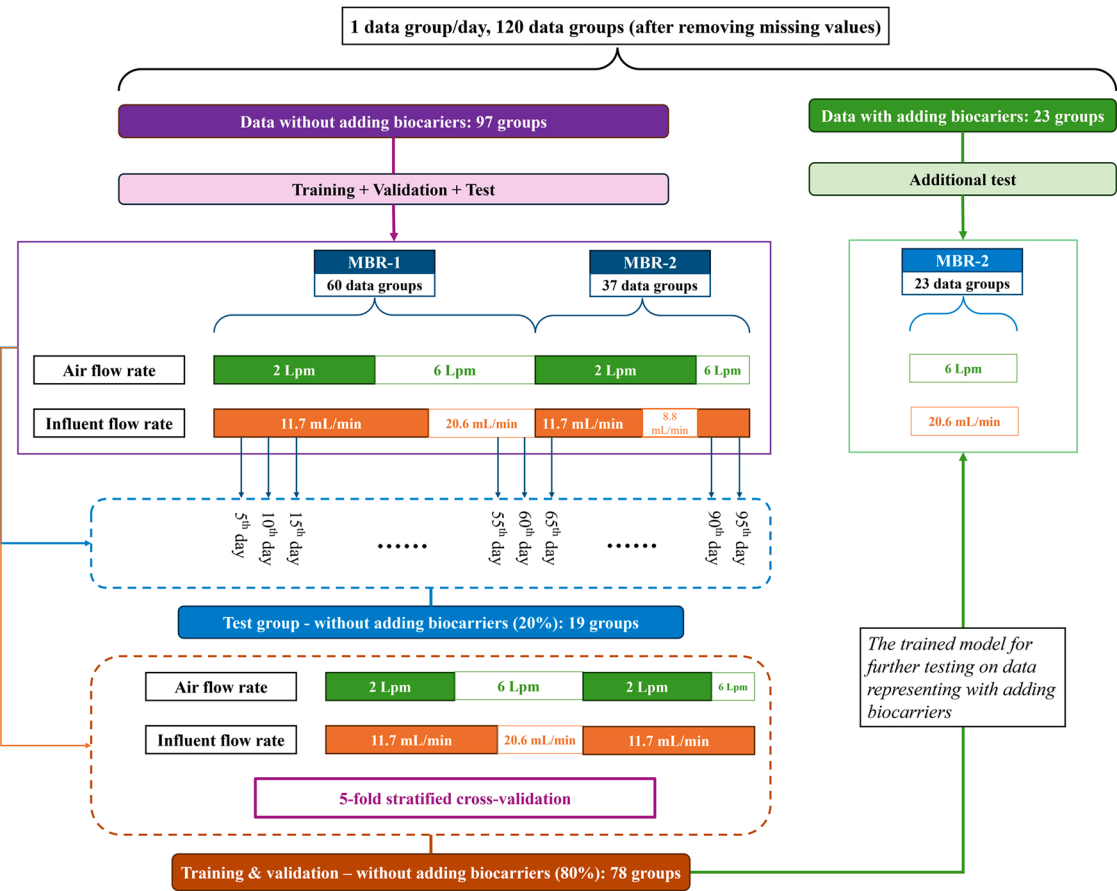


Fig. 3. Splitting strategy for training, validation and test groups.

3.2. Xác định các đặc trưng đầu vào

Previous studies have indicated that simpler models generally offer advantages for small datasets, including reduced overfitting risk, improved interpretability, and lower computational demands (Reynaert et al., 2023; Wang et al., 2025). For small, non-temporal datasets, complex structures such as artificial neural networks can increase the risk of overfitting (Chiroma et al., 2019).

Therefore, three interpretable algorithms with distinct bias–variance properties were selected. The logistic regression and decision tree-based models are normally cited as simulatable, i.e., interpretable (Breiman et al., 2017; Murdoch et al., 2019).

- **Logistic regression (LR)**: High bias, low variance. Applies logit transformation to linear regression for binary probability prediction, handling continuous/categorical variables with high interpretability and simple structure (Hosmer et al., 2013).
- **Random forest classification (RF)**: Intermediate bias-variance tradeoff. An ensemble method aggregating multiple decision trees via voting, improving accuracy and noise resistance for high-dimensional data (Breiman, 2001).
- **Extreme gradient boosting classification (XGB)**: Low bias, high variance. An optimized implementation of the gradient boosting algorithm with successive decision trees, each correcting the errors of its predecessor to minimize the loss function and improve predictive performance. Compared to traditional gradient boosting decision trees, XGB introduces enhancements such as regularization, parallel computing, and custom loss function definition (Chen and Guestrin, 2016).

All models were implemented using scikit-learn 1.0.2 (Python 3.9.13), with hyperparameters optimized via grid search.

2.2.7. Evaluation metrics and hyperparameter tuning criteria

The performance of the classification models was evaluated using a confusion matrix, contrasting predicted outcomes ("Sufficient"/"Insufficient") against actual conditions. The “Positive” label indicated sufficient nitrification, while the “Negative” label indicated insufficient nitrification, as defined in Section 2.1 Key metrics included true positives (TP: correctly classified sufficient nitrification), false positives (FP: erroneously classified sufficient under insufficient conditions), false negatives (FN: insufficient misclassifications despite actual sufficiency), and true negatives (TN: accurate classified as insufficient). Section 2.2.1

Minimizing the FPR was crucial for ensuring safe effluent water quality. For instance, in a control system integrated with this nitrification monitoring model, when insufficient nitrification occurs, more air should be supplied. However, if an FP occurs, the model incorrectly predicts sufficient nitrification, leading the control system to refrain from adding additional oxygen and unsafe effluent unsuitable for reuse. Meanwhile, a higher TPR was necessary to detect sufficient nitrification under conventional operation conditions, reducing operational costs by preventing unnecessary oxygen supply triggered by FN.

Besides, optimizing solely for low FPR could severely reduce TPR, as indicated in previous studies (Reynaert et al., 2023; Salvia et al., 2021; Yang et al., 2019). Therefore, a balanced approach considering both TPR (Eq. 1) and FPR (Eq. 2) was adopted, with precision (Eq. 3) serving as a key optimization metric. Precision reflects the proportion of TP predictions among all positive predictions. The TRP, FPR, and precision were calculated using the following equations:

$$TPR = \frac{TP}{TP + FN} \quad (1)$$

$$FPR = \frac{FP}{FP + TN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

Although precision was prioritized during hyperparameter tuning, accuracy, representing the proportion of correct predictions out of all cases, was also used for model evaluation in test groups:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Finally, the stability of model performance was evaluated on the test dataset by calculating 95 % confidence intervals for accuracy, precision, recall, and F1 score using 1000-times bootstrap resampling, following standard practices in machine learning validation.

2.2.8. Post hoc interpretability

Despite the inherent interpretability of decision tree-based methods, complex models necessitate post hoc interpretation (Murdoch et al., 2019).

Feature importance quantifies the contributions of input variables, aiding model’s interpretability improvement and guiding data collection strategies for enhanced performance. In LR, feature coefficients are derived by minimizing the loss function (negative log-likelihood) using optimization methods such as gradient descent (Hastie et al., 2009). They indicate the strength and direction of each feature’s linear relationship with the target variable (Murphy, 2012). For RF, feature importance is assessed by the reduction in impurity (e.g., Gini index) when a feature is used to split a node. Scores are averaged across all trees, capturing the overall contribution of each feature to the model (Liaw and Wiener, 2002). The feature importance in XGB is typically calculated using the Gain method. This method sums up and normalizes the gains from all splits involving a feature to compute its importance score (Chen and Guestrin, 2016).

SHAP analysis was implemented to leverage game theory to assign each feature an importance value for specific predictions. SHAP

3.3. Dự đoán trạng thái nitrat hóa khi không bổ sung giá thể sinh học

Table 1

Đánh giá thống kê về dự đoán trên nhóm kiểm tra khi có/không bổ sung giá thể sinh học cùng phân tích tầm quan trọng đặc trưng của mô hình đã huấn luyện bằng KDE (ước lượng mật độ nhân).

* “Validation (Ave)”: giá trị trung bình của các nhóm thăm định kiểm chứng chéo 5 lần phân tầng.

** Hệ số đặc trưng của LR được tính toán để đại diện cho tầm quan trọng đặc trưng trong dự đoán dựa trên hồi quy logistic.

Fig. 4. Phân tích SHAP: biểu đồ beeswarm (a: LR; b: RF; c: XGB) và biểu đồ thác nước (waterfall) (d: LR; e: RF; f: XGB).

Những nhận định sâu sắc hơn được rút ra từ các biểu đồ thác nước (waterfall). Trong mô hình LR (Fig. 4d), đầu ra chịu sự chi phối mạnh mẽ bởi một giá trị NH_4^+-N cao duy nhất (+2.84), bị triệt tiêu bởi các đóng góp âm từ NO_3^+-N và COD, phản ánh bản chất cộng tính của các mô hình tuyến tính. Sự phụ thuộc lớn vào một đặc trưng chi phối duy nhất như vậy làm nổi bật nguy cơ quá khớp: khi NH_4^+-N sai lệch khỏi các quy luật huấn luyện, mô hình gặp khó khăn trong việc khái quát hóa, giải thích cho sự suy giảm tỷ lệ dương tính thật (TPR) trong kiểm tra. Ngược lại, mô hình RF (Fig. 4e) thể hiện sự đóng góp cân bằng hơn từ nhiều đặc trưng, với các tác động có độ lớn thấp hơn nhưng phân tán, phù hợp với cơ chế lấy trung bình quyết định của mô hình tổng hợp (ensemble). Mô hình XGB (Fig. 4f) thể hiện tính phi tuyến và các hiệu ứng tương tác mạnh hơn. Tại đây, COD và NO_3^+-N cùng tác động tiêu cực chi phối, trong khi NH_4^+-N đóng góp tích cực nhưng không đủ để vượt qua các tín hiệu căng thẳng của quy trình, dẫn đến dự đoán âm tính. Mặc dù vậy, sự phức tạp của các tương tác đặc trưng phi tuyến này trong XGB, kết hợp với các mẫu huấn luyện bị giới hạn, đã làm gia tăng khả năng ghi nhớ nhiều đặc thù của tập dữ liệu, một dấu hiệu điển hình khác của hiện tượng quá khớp. Đáng chú ý, TMP và Air Flow Rate (lưu lượng sức khí) nổi lên như những yếu tố điều biến nhỏ nhưng không thể bỏ qua, cho thấy độ nhạy đối với các trạng thái vận hành.

values satisfy efficiency (prediction = base value + sum of feature contributions), symmetry, and additivity axioms, providing consistent local and global interpretability (Lundberg and Lee, 2017).

Furthermore, pairwise feature relationships were visualized via kernel density estimation (KDE) and pairwise feature plot. KDE could nonparametrically approximate continuous probability density functions by smoothing observed data points via symmetric kernel functions and bandwidth parameters (Scott, 2015). Pairwise feature plots, implemented as scatterplot matrices, reveal correlation and clustering patterns between features (Pekalska et al., 2005). It can help understand model classification performance by visualizing the confusion matrix (Waskom, 2021).

All analyses were implemented in Python 3.9.13 (Van Rossum and Drake, 2009).

3. Results analysis: data-driven model prediction

3.1. MBR performance and data collection

Phases 1–2 (no biocarriers) showed consistent COD (>85 % removal), NH₄⁺-N, and TN trends. Phase 1 effluent contained high NH₄⁺-N (14 ± 5 mg/L; 28 % removal) and TN (14 ± 2 mg/L; 23 % removal) with minimal NO₃⁻-N, indicating insufficient nitrification. Increasing aeration in Phase 2 improved NH₄⁺-N removal to 95 ± 1 % (effluent: 0.9 ± 0.2 mg/L). Meanwhile, the MLSS/MLVSS ratio stabilized at 0.92 ± 0.07 during Phase 1 and remained around 0.9 in Phase 2, indicating that most of the sludge consisted of organic fractions—a sign of healthy sludge conditions. In Phase 3, adding PVDF biocarriers to MBR-2 improved denitrification performance: TN removal reached 58 ± 21 %, compared to 34 ± 15 % in MBR-1 without biocarrier addition, while COD removal (90 ± 4 %) and NH₄⁺-N removal (94 ± 5 %) were maintained. The MBR-2 also exhibited reduced membrane fouling (TMP < 10 kPa) and higher biomass concentrations (MLSS: 2832 ± 831 mg/L; MLVSS: 2655 ± 806 mg/L) compared to the MBR-1 (MLSS: 2439 ± 648 mg/L; MLVSS: 2191 ± 543 mg/L). Overall, effluent COD and TN met EU Directive 91/271/EEC standards, though TN removal in Phase 3 was inconsistent, likely due to biofilm acclimatization. More detailed results are described in SI 2.

3.2. Input features determination

Following qualitative feature selection as shown in Section 2.2.2, nine input variables remained and were statistically evaluated. SI 3 identifies variables with significant correlations labeled with circles (p-value < 0.05) among the remaining input features. To simplify the model structure, only the influent flow rate was retained due to its similarity to the effluent flow rate, which demonstrated over 97 % flux recovery. Despite TMP’s weak correlation with nitrification, its inclusion could enhance model transferability by accounting for acyclic membrane fouling in long-term operations. Meanwhile, while the membrane cleaning record correlated with TMP, reflecting membrane maintenance needs, it was excluded to avoid material-specific biases. In addition, TN was omitted due to its marginal predictive contribution and impracticality for real-time monitoring. Although TN has certain correlation with the target output, TN sensors are costly and feature long measurement intervals (30–60 min, based on commercially available TN sensors), compared with sensors monitoring NO₃⁻-N and NH₄⁺-N.

Finally, six input features were selected for nitrification status prediction, including three operational factors (air flow rate, influent flow rate, and TMP) and three water quality variables (effluent COD, NO₃⁻-N, and NH₄⁺-N concentrations). This streamlined feature set balances model simplicity, accuracy, and field feasibility for nitrification monitoring.

3.3. Nitrification status prediction without biocarriers addition

Three classification models (LR, RF, XGB) were trained on datasets excluding biocarriers, with precision prioritized during hyperparameter optimization via stratified 5-fold cross-validation (SI 4). Table 1a summarizes prediction results using six selected input features, reporting accuracy, precision, TPR, and FPR for the test group, along with averaged results from cross-validation folds, illustrating the model’s predictive ability for nitrification status. Evaluation metrics for individual folds are provided in SI 5. Based on the limited data in the training (78) and test group (19), three key insights can be drawn from Table 1a:

- 1) LR, RF, and XGB achieved comparable test group performance (accuracy: 0.79–0.84). The confusion matrices for the test group predictions by each algorithm are provided in SI 6. This good fit could be attributed to the mechanistic relationship between the inputs and the target output, involving nitrogen conversion in the nitrification process (Huang et al., 2022; Omar et al., 2024).
- 2) With considering the superior performance in validation groups (accuracy: 0.90–0.91), RF and XGB underperformed LR in the test group, indicating potential overfitting risks with limited data.
- 3) Additionally, the small data limited the ability to fully evaluate the performance difference among algorithms with different complexity, hard to assess the influence of model complexity on TPR and FPR.

These results preliminarily support the effectiveness of the ML algorithm in predicting the nitrification status through TMP, air flow rate, influent flow rate, and effluent COD, NO₃⁻-N, and NH₄⁺-N concentrations.

However, compared with the prediction accuracy on the validation group, the decline of TPR in the test group can be attributed to several factors. One primary reason is the issue of class imbalance. In the training set, negative label samples accounted for 64.10 %, whereas in the test set, this proportion decreased to 57.89 %. Although this indicated a slightly lower proportion of negative Label samples in the test set, the imbalance remained significant. Such an imbalance biases the model toward the dominant class (negative

3.4. Thử nghiệm liên kịch bản

Các ma trận nhầm lẫn đầy đủ được cung cấp trong SI 9. Mặc dù về mặt lý thuyết, các kết quả này không thể so sánh trực tiếp với dữ liệu không có giá thể sinh học (Table 1a) do sự khác biệt về điều kiện vận hành và đặc tính của tập dữ liệu, những phát hiện này vẫn mang lại các hiểu biết vận hành về hành vi của mô hình liên điều kiện.

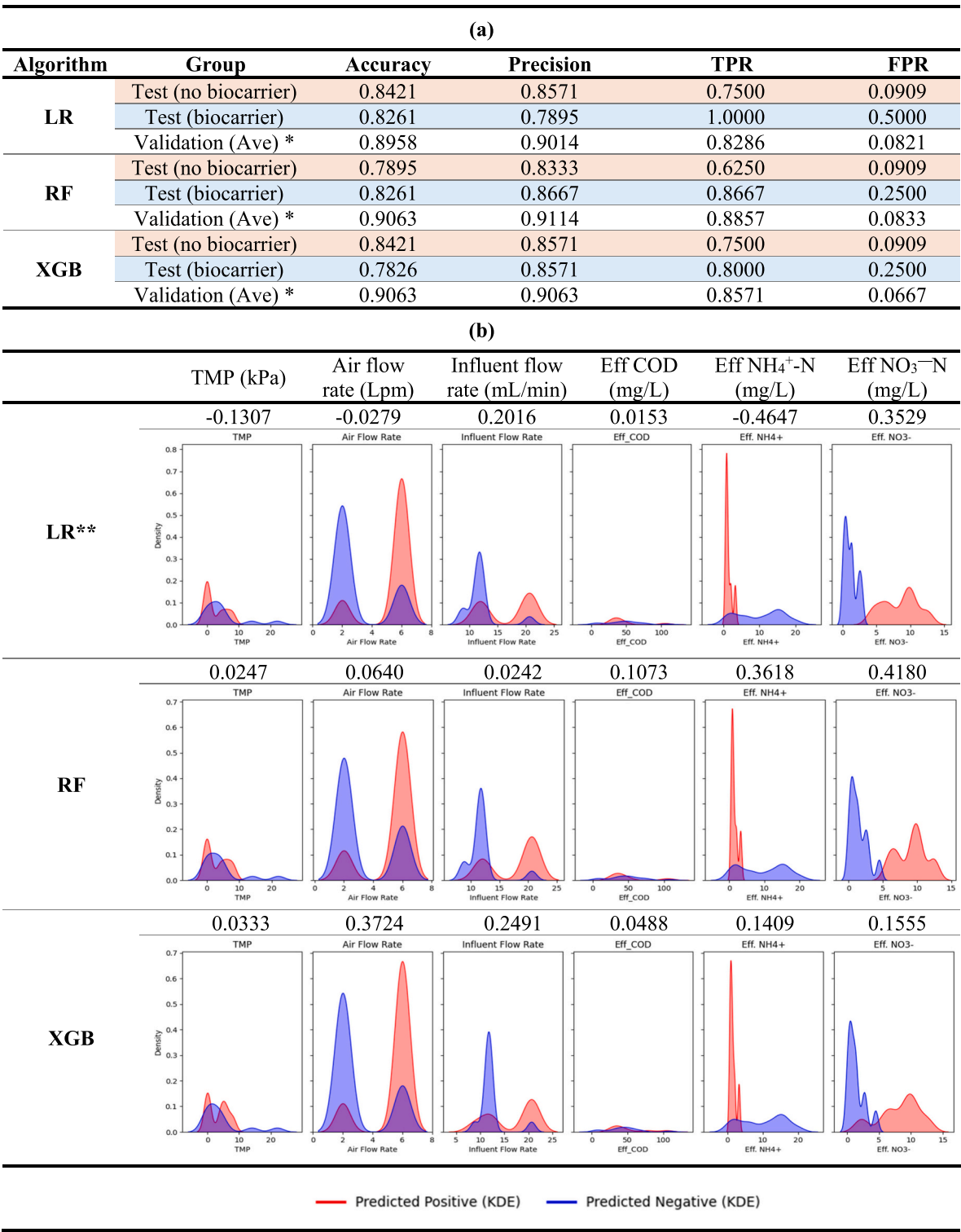
4. Thảo luận và triển vọng tương lai về vận hành bền vững

4.1. Giới hạn dữ liệu và thu thập dữ liệu

Bên cạnh hiện tượng quá khớp dự đoán do tập dữ liệu nhỏ hơn này như đã thảo luận trong Mục 3.3, các hạn chế về dữ liệu còn tác động đến các tiêu chí tối ưu hóa. Mặc dù việc tối đa hóa độ chuẩn xác (tỷ lệ dương tính thật (TPR) cao/tỷ lệ dương tính giả (FPR) thấp) đã được ưu tiên, dữ liệu hạn chế đã cản trở việc xác định ranh giới cho các đường cong tối ưu hóa TPR-FPR. Đường cong tối ưu hóa, như đã được chứng minh bởi Reynaert, et al. (2023), là một công cụ hữu ích để trực quan hóa quá trình hiệu chuẩn mô hình và định hướng việc lựa chọn tỷ lệ dương tính giả (FPR) phù hợp nhằm đánh giá hiệu năng trên các tập huấn luyện, thẩm định và kiểm tra. Vượt ra ngoài việc xác định tỷ lệ dương tính giả (FPR) thuần túy về mặt toán học, bối cảnh tái sử dụng nước đòi hỏi phải đánh giá rủi ro định lượng đối với các trường hợp dương tính giả (FP). Việc xác định các điểm vận hành hợp lý trước khi thử nghiệm liên kịch bản sẽ mang lại lợi ích cho quá trình thẩm định mô hình và điều chỉnh các điểm đặt (set-point). Đồng thời, các khoảng tin cậy rộng càng nhấn mạnh thêm sự cần thiết của các tập dữ liệu lớn hơn nhằm đánh giá toàn diện độ ổn định của mô hình và mức độ đóng góp của các đặc trưng. Sự phụ thuộc hiện tại vào NH₄⁺-N/NO₃⁻-N trong nước đầu ra cần được mở rộng để bao gồm các thông số vận hành, động học chất lượng nước và dữ liệu theo thời gian nhằm phục vụ cho phân tích độ nhạy đáng tin cậy.

Các cảm biến trực tuyến đóng vai trò thiết yếu cho việc thu thập dữ liệu nhất quán. Các đặc trưng tương thích với cảm biến (Mục 2.2.2) đã được ưu tiên trong nghiên cứu này nhằm tạo điều kiện thuận lợi cho việc thẩm định quy mô lớn trong tương lai. Đồng thời, việc tối ưu hóa tần suất đo lường phải cân bằng giữa các yêu cầu tái sử dụng nước và chi phí vận hành (Reynaert et al., 2024). Hơn nữa, việc xác định các giá trị đầu vào liên kết với các điểm phân loại sai, tăng tần suất đo lường và đánh giá kết quả dự đoán của mô hình trong những điều kiện như vậy có thể góp phần làm sáng tỏ các cơ chế sự cố và làm nền tảng cho việc nghiên cứu các hệ thống cảnh báo sớm.

Table 1
Statistical evaluation of predicting test group with/without adding biocarriers with feature importance analysis of trained model with kernel density estimates (KDE).



* “Validation (Ave)”: the average value of stratified 5-fold validation groups
** The feature coefficient of LR was calculated to represent the feature importance in logistics regression-based prediction.

Việc tích hợp các điều kiện vận hành nhiều thách thức hơn, chẳng hạn như các sự cố hệ thống (ví dụ: hỏng hóc sục khí hoặc hư hại màng), là điều cần thiết để nâng cao khả năng ứng dụng của mô hình trong các kịch bản xử lý thực tế (Reynaert, et al., 2023). Mặc dù nghiên cứu này đã bao gồm các điều kiện vận hành dưới mức tối ưu (ví dụ: nitrat hóa khi sục khí thấp), việc thu thập dữ liệu một cách có hệ thống trên các dạng sự cố khác nhau vẫn là điều thiết yếu.

4.2. Sự đánh đổi giữa các đặc trưng đầu vào đối với cấu trúc mô hình dựa trên dữ liệu

label), reducing its ability to accurately predict positive samples and resulting in TPR reduction in the test group. The problem was evident in all algorithms, where validation accuracy was even over 0.1 higher than that of the test group (Table 1a), highlighting the model’s susceptibility to overfitting to class proportions during the training phase due to the small dataset.

Moreover, feature distribution shifts exacerbated TPR declines and reflected underlying overfitting to training feature dependencies. In the training set, 58.67 % of the samples had an air flow rate of 2, and 41.33 % had an air flow rate of 6. In the test set, these proportions shifted to 52.63 % and 47.37 %, respectively. While this shift was relatively small, it could still influence the model’s generalization ability, particularly as the air flow rate is critical for predicting nitrification status. KDE plots (Table 1b) demonstrated that air flow rate distributions between positive and negative classes were well separated, which magnified the effect of even subtle distribution shifts. This explains why the RF and XGB models, although exhibiting high validation accuracy (0.9114 and 0.9063, respectively), suffered noticeable reductions in TPR during testing (0.6250 for RF and 0.7500 for XGB). Such discrepancies suggest that the models captured training-specific patterns too rigidly, a hallmark of overfitting.

Additionally, the scarcity of positive samples in the test set exacerbated overfitting risks. Limited training data forced models to overfit the dominant negative class, compromising their ability to generalize to unseen positive cases. Consequently, TPR declined during testing, as models struggled to correctly classify underrepresented positive samples. This phenomenon was particularly evident in RF, where precision remained relatively high (0.8333) but TPR declined sharply to 0.6250 in the test set. The discrepancy between validation and test outcomes demonstrated that RF overfit to negative-class features during training, compromising its predictive balance across both classes.

In contrast, the FPR remained stable across training and test sets. Such stability stemmed from the relatively uniform distribution of negative samples. As explained before, the training set’s high negative sample proportion enhanced model sensitivity to negative class

4.3. Cải thiện khả năng chuyển giao mô hình

Khả năng chuyển giao mô hình — năng lực của một mô hình trong việc thích ứng với các nhiệm vụ hoặc miền dữ liệu mới với yêu cầu tái huấn luyện tối thiểu — có thể được tối ưu hóa một cách có hệ thống thông qua các khung học chuyển giao có cấu trúc (Tian and Yu, 2023). Các chiến lược học chuyển giao bao gồm cân chỉnh đặc trưng, các mô hình tiền huấn luyện, kiến trúc siêu học tập (meta-learning), học tự giám sát, tăng cường dữ liệu và khả năng mở rộng của cấu trúc mô hình (Yu et al., 2022).

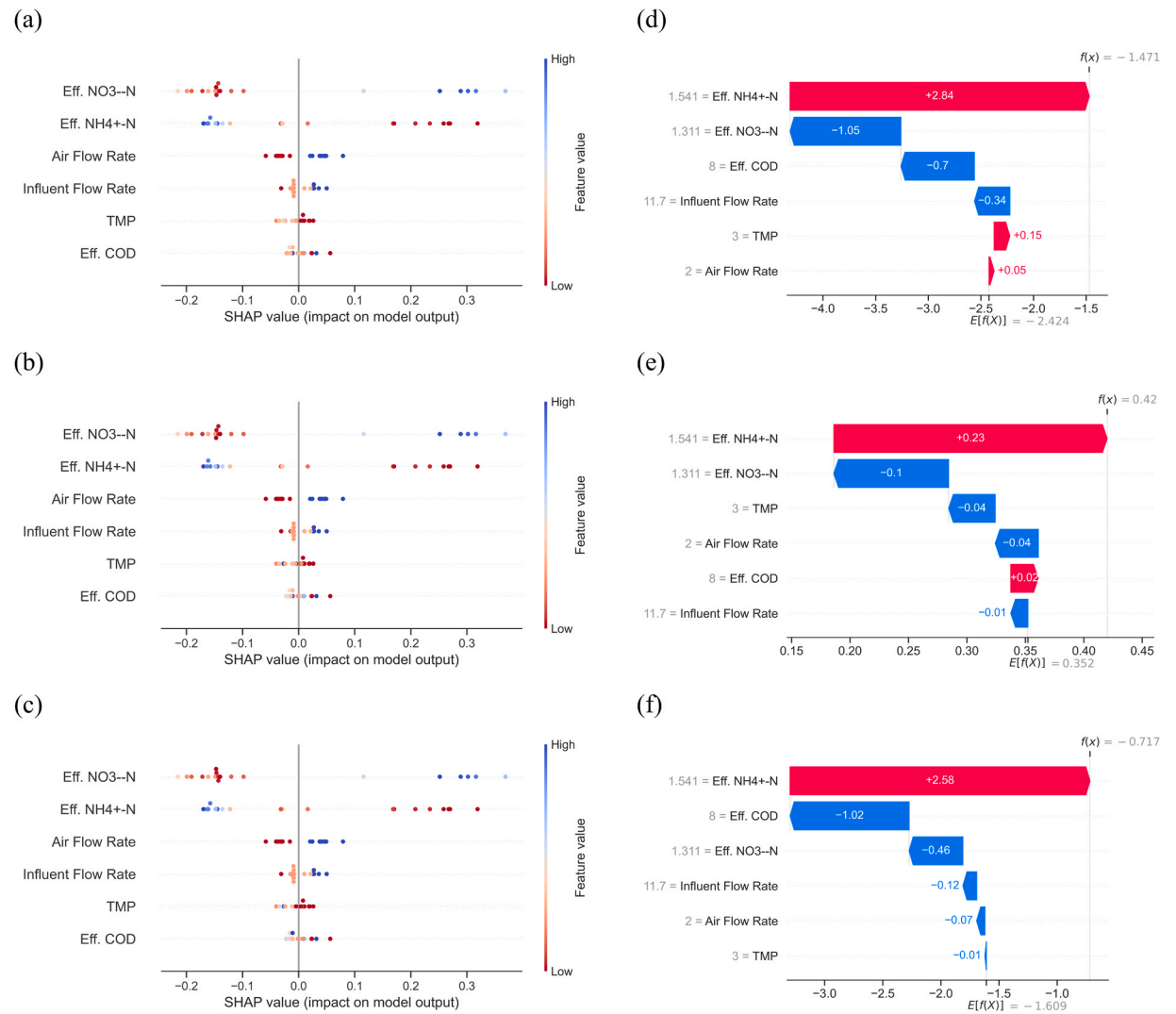


Fig. 4. SHAP analysis: beeswarm (a: LR; b: RF; c: XGB) and waterfall (d: LR; e: RF; f: XGB) plot.

detection. Therefore, the model maintained robust negative class identification when the test set’s negative sample ratio decreased slightly, sustaining low FPR levels. However, this apparent stability should not obscure the underlying problem: the divergence between high validation performance and degraded test performance across all three models (Table 1a) clearly demonstrated that overfitting was a systemic issue. The models effectively learned training-specific distributions rather than robust, generalizable patterns required for reliable deployment.

In summary, the decline in TPR in the test set was primarily due to class imbalance, feature distribution shifts, and the limited number of “positive” label samples, which increased the overfitting risk. Conversely, the stability of FPR could be attributed to the consistent distribution of negative samples across the training and test sets. These results underscore the challenges in ensuring model robustness under data distribution shifts — a critical consideration for deploying ML in dynamic wastewater treatment environments.

Feature importance analysis elucidated prediction mechanisms and guided data collection strategies. Table 1b shows importance scores with Kernel Density Estimates (KDE) for LR, RF, and XGB models. In LR, regression coefficients indicated feature-prediction directionality/magnitude. For RF and XGB, importance scores quantified feature contributions. KDE visualized probability density distributions for Predicted Positive (red) and Predicted Negative (blue) classes. Curve peaks revealed value concentration regions (e.g., Eff_COD peaked at ~30 for Positive class). Class discriminative ability was assessed through KDE overlap: minimal overlap indicated strong separability (e.g., Air Flow Rate showed moderate separation). Curve width reflected feature variance—narrow curves denoted consistent class values, while broader curves indicated higher variability.

Feature importance and KDE analysis confirmed effluent $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ as primary predictors of nitrification status. Although mechanistically relevant, influent flow rate (11.7–20.6 ML/min) and air flow rate (2–6 Lpm) showed limited contributions due to low operational variability. Effluent COD likewise had a minor impact, evidenced by its lower coefficient of variance (55.25 %) versus $\text{NH}_4^+\text{-N}$ (93.82 %) and $\text{NO}_3^-\text{-N}$ (101.72 %)—calculated as standard deviation relative to mean. This variability disparity explained differential feature importance.

SI 7 and 8 display pairwise plots across models and feature importance quantification across validation groups respectively, which confirmed effluent $\text{NO}_3^-\text{-N}/\text{NH}_4^+\text{-N}$ as primary predictors, aligning with their biochemical centrality in nitrification. However, XGB validation groups showed unexpected prominence of stable influent and airflow rates over N-species. This stemmed from XGB’s boosting mechanism: sequential trees prioritize features correcting predecessor errors, dynamically weighting features during incremental optimization (Bentéjac et al., 2021; Nguyen et al., 2024). Such adaptability could enhance model flexibility across data groups but may inflate transient feature importance and overfitting risks.

Fig. 4 illustrates the SHAP analysis results for three predictive models—LR, RF, and XGB—including global importance (beeswarm plots, Fig. 4a–c) and instance-level decomposition (waterfall plots, Fig. 4d–f). The beeswarm plots revealed consistent feature importance rankings across all models: Effluent $\text{NH}_4^+\text{-N}$ and Effluent $\text{NO}_3^-\text{-N}$ dominate, followed by Effluent COD and TMP, suggesting robust, model-agnostic relevance of nitrogen and organic matter indicators in treatment performance prediction, consistent with the results of feature importance analysis.

More nuanced insights arose from the waterfall plots. In the LR model (Fig. 4d), the output was strongly driven by a single high $\text{NH}_4^+\text{-N}$ value (+2.84), counteracted by negative contributions from $\text{NO}_3^-\text{-N}$ and COD, reflecting the additive nature of linear models. Such strong reliance on a single dominant feature highlighted an overfitting risk: when $\text{NH}_4^+\text{-N}$ deviates from training patterns, the model struggles to generalize, explaining the decline in test TPR. In contrast, the RF model (Fig. 4e) showed a more balanced contribution from multiple features, with lower-magnitude but distributed impacts, consistent with ensemble decision averaging. The XGB model (Fig. 4f) exhibited stronger nonlinearity and interaction effects. Here, COD and $\text{NO}_3^-\text{-N}$ jointly exerted dominant negative influence, while $\text{NH}_4^+\text{-N}$ contributed positively but was insufficient to overcome process stress signals, resulting in a negative prediction. Nevertheless, the complexity of these nonlinear feature interactions in XGB, combined with limited training samples, raised the likelihood of memorizing dataset-specific noise, another hallmark of overfitting. Notably, TMP and Air Flow Rate emerged as minor but non-negligible modulators, indicating sensitivity to operational states.

Furthermore, combined with the analysis of the effluent concentration of $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$, which had the main influence on the prediction, and by tracing back the characteristics in the data set, it was found that false positives mainly occurred when the effluent $\text{NO}_3^-\text{-N}$ was 9 mg/L and $\text{NH}_4^+\text{-N}$ was 2 mg/L. That is, the effluent quality showed a relatively obvious sufficient nitrification potential, but the reactor was in an insufficient nitrification state simultaneously. Compared with the output labels in the training dataset, this was mainly because when the effluent $\text{NO}_3^-\text{-N}$ concentration was greater than 9 mg/L and the $\text{NH}_4^+\text{-N}$ concentration was less than 2 mg/L, the data labels were all positive, that is, sufficient nitrification, which in turn affected the prediction in the test set. This phenomenon indicated that the models overfitted the label distribution patterns in the training set and thus misclassified borderline cases in the test set, exposing a lack of generalization.

The SHAP analysis revealed that effluent $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ were the most influential predictors across all models, underscoring their critical role as indicators of nitrification performance in MBR systems. This suggested that real-time monitoring of these nitrogen species—rather than relying only on conventional parameters like TMP, DO in the reactor—could enhance operational control while monitoring essential effluent water quality parameters. Specifically, the identification of borderline cases (e.g., simultaneous $\text{NH}_4^+\text{-N} \approx 2$ mg/L and $\text{NO}_3^-\text{-N} \approx 9$ mg/L where false positives occur) highlighted process instability regions that were prone to misclassification. Such thresholds could inform early-warning systems: when nitrogen levels approach these critical values, the intelligent control system should proactively adjust aeration rates (the “Air Flow Rate” SHAP contributions support its modulating role) or check sludge retention time to prevent nitrification failure. Moreover, multi-parameter synergistic management is essential to maintain robust nitrification and avoid model overfitting to transient noise. These insights bridge data-driven predictions with actionable process strategies, promoting adaptive control in MBR operations.

Therefore, based on this proof of concept, further experiments should be designed to investigate more variable influent water

Tuyên bố đóng góp của tác giả theo CRediT

Cách tiếp cận đa chiều này giải quyết các hạn chế hiện tại về khả năng tổng quát hóa của mô hình, đồng thời thích ứng với tính biến thiên vốn có trong xử lý nước về quy mô vận hành, đặc tính nước đầu vào và cấu hình quy trình. Các nghiên cứu trong tương lai cần ưu tiên việc thiết lập các tập dữ liệu đối chuẩn mở nhằm chuẩn hóa việc đánh giá khả năng chuyển giao trên các kịch bản đa dạng.

4.4. Kiểm soát bền vững dựa trên dữ liệu

Sau khi huấn luyện mô hình với các đặc trưng đầu vào phù hợp và lượng dữ liệu đầy đủ được thu thập một cách có hệ thống (như đã thảo luận trong Mục 5.1 và 5.2), một hệ thống kiểm soát tự động có thể được phát triển thêm để vận hành thông minh hệ thống MBR. Fig. 5 trình bày lưu đồ kiểm soát dựa trên mô hình dựa trên dữ liệu nhằm quản lý tự động quy trình xử lý của MBR. Để đơn giản hóa phần giải thích, mô hình giả định các điều chỉnh lưu lượng sục khí dưới điều kiện lưu lượng dòng vào cố định (nghĩa là HRT không đổi). Trước khi triển khai hệ thống kiểm soát dựa trên dữ liệu, giá trị cơ sở của lưu lượng sục khí (a_i) và đơn vị gia tăng (A) cần được xác định bằng cách phân tích các đặc tính của nước xám đầu vào trong một khoảng thời gian cụ thể. Tần suất đo cần được điều chỉnh phù hợp với các phương án tái sử dụng và các yêu cầu chất lượng nước đầu ra tương ứng. Các tiêu chí lựa chọn những yếu tố vận hành này cần được nghiên cứu thêm.

Fig. 5. Lưu đồ kiểm soát MBR dựa trên mô hình dự đoán phân loại. * Lưu đồ thể hiện cơ chế kiểm soát dưới điều kiện lưu lượng dòng vào cố định. * Hình chữ nhật nét đứt màu xám biểu thị một lần dự đoán và thao tác kiểm soát tương ứng. * “ a_i ” là giá trị cơ sở của lưu lượng sục khí, “A” là đơn vị tăng lưu lượng sục khí. * Hình chữ nhật màu đỏ biểu thị lưu lượng sục khí là a_i L/min; hình chữ nhật màu xanh lá cây biểu thị lưu lượng sục khí là $(a_i + A)$ L/min; hình chữ nhật màu xanh dương biểu thị lưu lượng sục khí là $(a_i + 2A)$ L/min; hình chữ nhật màu tím biểu thị lưu lượng sục khí là $(a_i + 3A)$ L/min; hình chữ nhật màu nâu biểu thị lưu lượng sục khí là $(a_i + 4A)$ L/min (các hình chữ nhật có cùng màu biểu thị cùng một mức lưu lượng sục khí).

Lưu lượng sục khí có thể được tăng hoặc giảm để duy trì kiểm soát bền vững mà không tiêu tốn quá nhiều năng lượng. Giả định rằng hệ thống vận hành ở mức a_i và trạng thái nitrat hóa được dự đoán tại thời điểm t_i (Fig. 5). Nếu mô hình dự đoán nitrat hóa hoàn toàn (nhãn “True”), lưu lượng sục khí sẽ duy trì ở mức a_i ; ngược lại, lưu lượng sẽ tăng thêm một lượng A lên mức $a_i + A$. Sau đó, nếu hệ thống vận hành qua một khoảng đo với lưu lượng sục khí là $a_i + A$, tại khoảng đo tiếp theo t_{i+1} , lưu lượng sục khí sẽ quay trở lại giá trị cơ sở a_i nếu dự đoán là nitrat hóa hoàn toàn; nếu không hoàn toàn, lưu lượng sẽ tăng lên $a_i + 2A$. Quá trình này lặp lại ở mỗi khoảng thời gian tiếp theo (t_{i+2} , t_{i+3} , v.v.), điều chỉnh lưu lượng sục khí một cách tương ứng. Như thể hiện trong Fig. 5, sau khi xác định xem hệ thống xử lý đã nitrat hóa hoàn toàn hay chưa, lưu lượng sục khí sẽ tăng hoặc giảm theo từng đơn vị A. Lưu lượng sục khí có thể được giảm xuống mức tối thiểu là giá trị cơ sở a_i , nhưng nếu vượt quá công suất tối đa của thiết bị sục khí, một hệ thống cảnh báo sớm cần kích hoạt sự can thiệp của con người, chẳng hạn như xả bỏ nước xám hoặc bổ sung thiết bị sục khí. Hơn nữa, tần suất giám sát có thể được tăng lên trong các giai đoạn sử dụng nước cao điểm nhằm nâng cao hiệu quả xử lý và độ an toàn tái sử dụng.

Khung phân loại nhị phân chứng minh tính khả thi trong vận hành dưới các điều kiện hạn chế về dữ liệu. Việc mở rộng khung hiện tại sang các trạng thái nitrat hóa đa lớp—chẳng hạn như một phương án dự đoán bốn mức (25 %, 50 %, 75 % và nitrat hóa hoàn toàn)—có thể cho phép kiểm soát sục khí theo tỷ lệ. Sự cải tiến này sẽ tối ưu hóa việc sử dụng năng lượng, giảm đầu chân carbon và giúp đảm bảo chất lượng nước đầu ra ổn định. Những mô hình như vậy đòi hỏi các tập dữ liệu mở rộng từ các cảm biến trực tuyến (Mục 4.1) và các kỹ thuật tối ưu hóa chuyên biệt: tinh chỉnh ngưỡng đa lớp nhằm giải quyết chi phí phân loại sai, cùng các phương pháp dành cho dữ liệu mất cân bằng (Davis and Goadrich, 2006; Haibo He and Garcia, 2009; Saito and Rehmsmeier, 2015). Việc lựa chọn ngưỡng cần tính đến chi phí và rủi ro của phân loại sai, vượt ra ngoài các cân nhắc thuần túy về mặt toán học. Ngoài ra, định nghĩa các lớp cần phải phù hợp với các đặc tính của dòng vào và các yêu cầu tái sử dụng.

Việc tích hợp mô hình phân loại đa lớp vào hệ thống kiểm soát cho phép các điều chỉnh chính xác hơn, có khả năng loại bỏ nhu cầu thay đổi theo từng bước gia tăng (chẳng hạn như A trong Fig. 5), tuy nhiên các chiến lược giảm sục khí tối ưu sau khi dự đoán nitrat hóa hoàn toàn—giảm dần dần so với giảm ngay lập tức—đòi hỏi phải được kiểm chứng trong phòng thí nghiệm/quy mô thử nghiệm (pilot). Giảm dần dần có thể phù hợp hơn để duy trì hiệu quả đối với các tiêu chuẩn tái sử dụng nghiêm ngặt, đòi hỏi phải có các đánh giá rủi ro - chi phí.

5. Kết luận

quality based on the range of gray water quality characteristics summarized in existing literature to simulate the highly variable characteristics of gray water quality. Meanwhile, collecting different influent flow rates, that is, the water quality variation under different HRTs, would allow further analysis of the lag relationship between the effluent ammonia nitrogen and nitrate nitrogen concentrations, and the reaction state in the reactor. Furthermore, designing time series data collection, such as collecting and measuring water samples every hour, could help determine the time series relationship between the effluent water quality changes and the reaction state in the reactor at the same time from a more dynamic perspective. In summary, broader datasets with more dynamics are necessary to further test the feasibility of using effluent quality to predict or even control the reaction state inside the reactor. Such expanded and diversified datasets are also critical to mitigate current overfitting risks and improve the models’ robustness for real-world applications. [Sections 4.1 and 4.2](#) further discuss the impact of data limitation and input feature on the existing model framework.

3.4. Cross-scenario test

Building upon the demonstrated efficacy of the model in conventional nitrification systems ([Section 3.3](#)), its cross-scenario prediction capability was further evaluated in MBRs incorporating biocarriers for enhanced simultaneous nitrification-denitrification. The base model trained on no-biocarrier operations was directly tested to predict nitrification status in biocarrier-augmented systems without retraining.

As shown in [Table 1a](#), the cross-condition testing yielded three principal observations:

- 1) Compared with prediction performance of the test group without biocarriers, the RF model exhibited improvement in both accuracy ($0.79 \rightarrow 0.83$) and precision ($0.83 \rightarrow 0.87$) under biocarrier-amended conditions, suggesting superior adaptability to simultaneous nitrification-denitrification system complexities.
- 2) TPRs of all these three algorithms were greater than or equal to 0.8, especially that of LR, which reached 1, showing a significant increase compared with the test group without adding biological carriers.
- 3) Corresponding FPR values rose significantly, particularly for LR (FPR=0.50), indicating only 50 % negative sample identification accuracy.

Full confusion matrices are provided in SI 9. While these results are theoretically not directly comparable to those from non-biocarriers data ([Table 1a](#)) due to differences in operating conditions and dataset characteristics, these findings provide operational insights into cross-condition model behavior.

The comparatively high TPR and FPR observed in the test group of the MBR with biocarrier addition can be explained by the following reasons. First, feature consistency in the test set played a significant role. In the training set, the input feature "air flow rate" included two values (2 and 6 Lpm), whereas all samples in the test set had an air flow rate of 6 Lpm. This consistency simplified the model’s decision rules, reducing the likelihood of misclassifying negative label samples (as shown in the KDEs in [Table 1b](#)). Second, the limitations of the decision boundary also contributed to the results. Despite effective positive classification at 6 Lpm, the model retained residual misclassification tendencies for negative samples. This effect was amplified by the test set’s limited negative sample size ($n = 8$), rendering FPR measurements particularly sensitive to individual misclassifications.

Finally, bootstrap sampling ($n = 1000$) revealed that models without biocarriers exhibited high precision/accuracy but wide 95 % confidence intervals (indicating instability), whereas biocarrier addition narrowed confidence intervals—particularly for RF, which achieved optimal stability and performance (detailed confidence interval analysis for both scenarios is provided in SI 10).

4. Discussion and future perspective of sustainable operation

4.1. Data limitation and data collection

Data scarcity remains a fundamental constraint in developing ML models for environmental applications, particularly when adapting pre-trained models to novel operational scenarios through transfer learning ([Dansawad et al., 2023](#); [Imen et al., 2023](#); [Lowe et al., 2022](#); [Zhang, 2022](#); [Zhu et al., 2022](#)). This dataset—collected from experimental operations instead of being designed for ML development—limited model evaluation despite > 0.8 test accuracy. Larger datasets spanning diverse operation conditions are essential for robust decision boundaries, particularly when distinguishing transient process states during simultaneous nitrification-denitrification.

In addition to the prediction overfitting due to this smaller dataset discussed in [Section 3.3](#), data constraints impact optimization criteria. Although precision maximization (high TPR/low FPR) was prioritized, limited data hindered boundary definition for TPR-FPR optimization curves. The optimization curve, as demonstrated by [Reynaert, et al. \(2023\)](#), is a useful tool for visualizing the model calibration process and guiding the selection of an appropriate FPR to evaluate performance across the training, validation, and test sets. Beyond mathematical FPR determination, water reuse contexts require quantitative risk assessment of FPs. Identifying reasonable operating points before cross-scenario testing would benefit model validation and set-point adjustments. Meanwhile, wide confidence intervals further underscore the need for larger datasets to evaluate model stability and feature contributions comprehensively. Current reliance on effluent $\text{NH}_4^+\text{-N}/\text{NO}_3^-\text{-N}$ should expand to include operational parameters, water quality dynamics, and temporal data for credible sensitivity analysis.

Online sensors are vital for consistent data collection. Sensor-compatible features ([Section 2.2.2](#)) were prioritized in this research to

Siyuan Wang: Viết – rà soát & chỉnh sửa, Viết – bản thảo gốc, Trực quan hóa, Phần mềm, Phương pháp luận, Điều tra, Phân tích hình thức, Lên ý tưởng. Duc-Viet Nguyen: Viết – rà soát & chỉnh sửa, Điều tra. Seow Wah How: Phương pháp luận, Điều tra, Phân tích hình thức, Quản lý dữ liệu. Di Wu: Viết – rà soát & chỉnh sửa, Giám sát, Điều tra, Huy động tài trợ, Lên ý tưởng.

Trong quá trình chuẩn bị công trình này, (các) tác giả đã sử dụng DEEPSEEK™ nhằm cải thiện tính dễ đọc và ngôn ngữ của bản thảo. Sau khi sử dụng công cụ/dịch vụ này, các tác giả đã xem xét và chỉnh sửa nội dung khi cần thiết, đồng thời chịu toàn bộ trách nhiệm về nội dung của bài báo đã xuất bản.

Tuyên bố về xung đột lợi ích

Lời cảm ơn

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Phụ lục A. Thông tin bổ trợ

Dữ liệu bổ trợ liên kết với bài báo này có thể được tìm thấy trong phiên bản trực tuyến tại doi:10.1016/j.eti.2025.104541.

Tính sẵn có của dữ liệu

Tài liệu tham khảo

Allen, G.I., Gan, L., Zheng, L., 2024. Interpretable machine learning for discovery: statistical challenges and opportunities. *Annu. Rev. Stat. Appl.* 11 (1), 97–121. <https://doi.org/10.1146/annurev-statistics-040120-030919>.

facilitate future large-scale validation. Meanwhile, optimizing measurement frequency must balance water reuse requirements and operational costs (Reynaert et al., 2024). Furthermore, identifying the input values linked to misclassification points, increasing measurement frequency, and evaluating the model prediction results in such conditions could contribute to elucidating failure mechanisms and underpinning early-warning systems investigation.

Current model validation using synthetic greywater provides proof-of-concept but requires extension to real wastewater matrices. To address this, future work should implement a phased framework:

- Phase I: Synthetic → Synthetic (Current study)
- Phase II: Synthetic → Real (Next validation step)
- Phase III: Hybrid training (e.g., 60 % real + 40 % synthetic)

Incorporating more challenging operational conditions, such as system failures (e.g., aeration malfunction or membrane damage), is essential for improving the model’s applicability in real-world treatment scenarios (Reynaert, et al., 2023). Though this study included suboptimal operations (e.g., low-aeration nitrification), systematic data collection across failure modes remains essential.

4.2. Trade-offs of input features for data-driven model structure

The model uses effluent quality to assess concurrent nitrification status, effectively employing historical data to predict current conditions. However, influent flow variations affecting hydraulic retention time (HRT) introduce input-output time lags that may compromise accuracy. Accounting for inflow dynamics is thus critical, especially with highly variable greywater generation (Pinto and Maheshwari, 2015). Future data collection should address flow variability to enhance predictions, potentially through flow-specific sub-models. Due to extensive DO data gaps, air flow rate served as the sole oxygen proxy. However, observed DO heterogeneity—driven by fluid dynamics, uneven oxygen transfer, microbial activity gradients, and reactor design (Gillot and Héduit, 2000)—necessitates distributed DO sensors for accurate future modeling.

These findings underscore the need for strategic input feature selection. Features must balance statistical correlations with practical relevance, as imbalanced datasets can distort input-output relationships. Systematic comparison of feature combinations is essential, noting that added inputs do not guarantee accuracy gains despite correlations (Linardatos et al., 2020; Reynaert et al., 2023). Besides, feature stability and reliability are equally critical: water-quality-only inputs risk long-term bias without rigorous sensor maintenance. Measurement errors and recalibration schedules must be integrated into robust model design. Finally, sensor costs should be factored into monitoring systems to enhance real-world applicability and shorten payback periods.

4.3. Model transferability improvement

Model transferability — the ability of a model to adapt to new tasks or domains with minimal retraining — can be systematically optimized through structured transfer learning frameworks (Tian and Yu, 2023). Transfer learning strategies include feature alignment, pre-training models, meta-learning architectures, self-supervised learning, data augmentation, and scalability of model structure (Yu et al., 2022).

Despite broad industrial investigations and applications of transfer learning, its potential for cross-scenario adaptation in water treatment systems remains underexplored. Current interdisciplinary advances demonstrate promising pathways: Li et al. (2021) improved the short-term traffic prediction accuracy via fine-tuning pre-trained models from data-rich areas; Zhou et al. (2025) enhanced battery performance transferability through condition-robust data augmentation; Nieves-Avendano et al. (2023) successfully adapted a pre-trained MobileNetV2 model for mechanical fault detection via targeted domain fine-tuning.

Building on the cross-scenario test (Section 3.4), three enhancement pathways are proposed for enhancing transferability of water treatment models:

- Data-centric optimization: systematic experimental designs should prioritize wide operational condition coverage and representative data distributions during collection. Data augmentation efficacy directly depends on baseline dataset diversity
- Pre-trained frameworks: development of domain-specific pre-trained models requires cross-institutional data sharing, particularly for systems handling varied water qualities or scale-dependent processes. Input features should explicitly incorporate treatment facility scale parameters due to their hydraulic and biochemical implications.
- Model structure innovation: Clustered sub-models using feature grouping and range-partitioned training data could further improve local accuracy while maintaining overall system adaptability. Further attempts can also be made in the innovative application of algorithms (e.g., meta learning).

This multipronged approach addresses current limitations in model generalization while aligning with water treatment’s inherent variability in operational scales, influent characteristics, and process configurations. Future work should prioritize establishing open benchmarking datasets to standardize transferability evaluation across diverse scenarios.

4.4. Data-driven sustainable control

After training the model with suitable input features and sufficient data collected systematically (as discussed in sections 5.1 and

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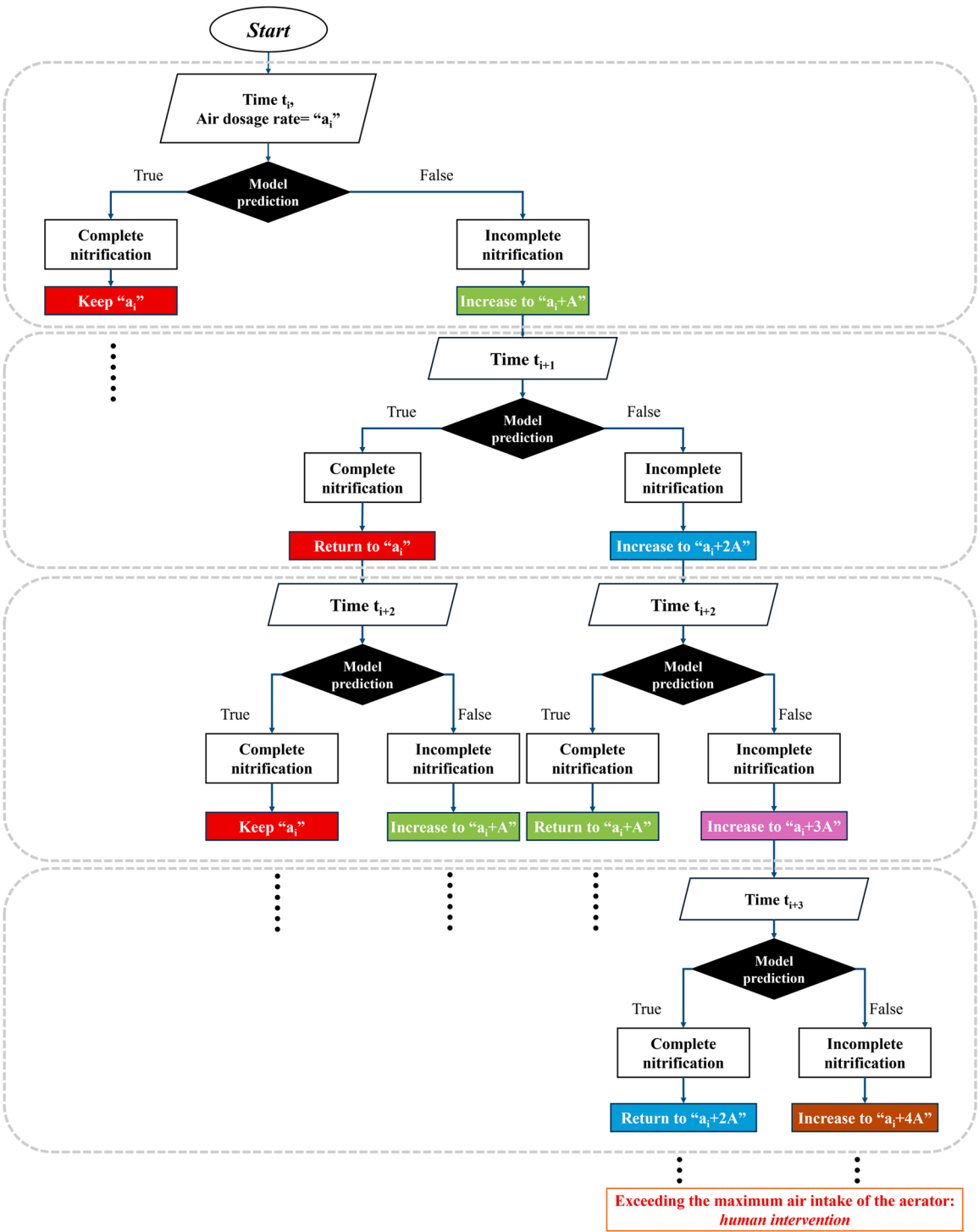


Fig. 5. Flow chart of MBR control based on classification prediction model. * The flow chart presents the control mechanism under a fixed influent flow rate. * The grey dashed rectangle represents one prediction and the corresponding control operation. * “ a_i ” is the baseline value of air flow rate, “ A ” is the unit of air flow rate increase. * The red rectangle means that the air flow rate is a_i L/min; the green rectangle means that the air flow rate is $(a_i + A)$ L/min; the blue rectangle means that the air flow rate is $(a_i + 2A)$ L/min; the purple rectangle means that the air flow rate is $(a_i + 3A)$ L/min; the brown rectangle means that the air flow rate is $(a_i + 4A)$ L/min (rectangles of the same color represent the same air flow rate).

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5.2), an automatic control system could be further developed for intelligent operation of MBR system. Fig. 5 presents a control flowchart based on the data-driven model for automatically managing the MBR treatment process. To simplify the explanation, it assumes air flow rate adjustments under a fixed influent flow rate (i.e., constant HRT). Before implementing the data-driven control system, the baseline air flow rate (a_i), and the increment unit (A) should be determined by analyzing influent greywater characteristics over a specific period. The measurement frequency should be tailored to reuse options and corresponding effluent quality requirements. The criteria for selecting these operational factors require further investigation.

The air flow rate can be increased or decreased to maintain sustainable control without excessive energy use. Assumably, the system operates at a_i , and nitrification is predicted at time t_i (Fig. 5). If the model predicts complete nitrification ("True" label), the air flow rate remains at a_i ; otherwise, it is increased by A to $a_i + A$. Then, if operating the system through a measurement interval with air flow rate of $a_i + A$, At the next measurement interval t_{i+1} . The air flow rate returns to the baseline value a_i if complete nitrification is predicted; if incomplete, it increases to $a_i + 2A$. This process repeats at each subsequent interval (t_{i+2} , t_{i+3} , etc.), adjusting the air flow rate accordingly. As shown in Fig. 5, after determining whether the treatment system is fully nitrified, the air flow rate increases or decreases in units of A . The air flow rate can be reduced to the baseline a_i minimally, but if it exceeds the aerator's maximum capacity, an early warning system should trigger human intervention, such as discarding greywater or adding aeration equipment. Moreover, monitoring frequency can be increased during peak water usage to enhance treatment efficiency and reuse safety.

The binary classification framework demonstrates operational feasibility under data-constrained conditions. Extending the current framework to multiclass nitrification states—such as a four-level prediction scheme (25 %, 50 %, 75 %, and complete nitrification)—could enable proportional aeration control. This refinement would optimize energy use, reduce carbon footprint, and help ensure consistent effluent quality. Such models require expanded datasets from online sensors (Section 4.1) and specialized optimization techniques: multi-class threshold tuning addressing misclassification costs, and methods for imbalanced data (Davis and Goadrich, 2006; Haibo He and Garcia, 2009; Saito and Rehmsmeier, 2015). Threshold selection should account for the cost and risk of misclassification, beyond purely mathematical considerations. Additionally, class definitions should align with influent characteristics and reuse requirements.

Integrating the multiclassification model into the control system enables more precise adjustments, potentially eliminating the need for incremental changes (such as A in Fig. 5), but optimal aeration reduction strategies post-complete nitrification prediction—gradual vs. immediate—require lab/pilot validation. Gradual reduction may be better for maintaining efficiency for strict reuse standards, necessitating risk-cost evaluations.

5. Conclusions

This work demonstrates the viability of interpretable ML-based models (LR, RF, XGB) for monitoring MBR nitrification by using six sensor-compatible features: airflow, influent flow, TMP, and effluent concentrations of COD, NO₃⁻-N, and NH₄⁺-N. Strategic feature selection enabled robust classification (precision >0.85 for LR/XGB). RF excelled in cross-scenario testing (precision=0.87) for predicting biocarrier operations using training data from non-biocarrier systems, despite a decline in FPR due to feature distribution shifts. All models maintained TPR ≥ 0.80 and accuracy > 0.75 in cross-scenario testing, though limited data constrained stability. Results emphasized ML's potential to enhance MBR control by balancing precision (minimizing FPs) with transferability across configurations. Post hoc analysis (feature importance, KDE, pairwise plots, and SHAP analysis) confirmed effluent NO₃⁻-N and NH₄⁺-N as dominant predictors. Wide 95 % confidence intervals reflected instability from small dataset scope.

Future research should prioritize systematic data acquisition and model training across diverse influent water quality and DO regimes. Concurrently, expanding boundary conditions—specifically targeted collection under high NO₃⁻-N/low NH₄⁺-N conditions is essential to optimize model architecture. This work establishes a framework for deploying interpretable data-driven models in decentralized greywater reuse, advancing intelligent MBR control while addressing transferability barriers in real-world implementations.

CRediT authorship contribution statement

Si yuan Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Duc-Viet Nguyen:** Writing – review & editing, Investigation. **How Seow-Wah:** Methodology, Investigation, Formal analysis, Data curation. **WU Di:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used DEEPSEEK™ in order to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.eti.2025.104541](https://doi.org/10.1016/j.eti.2025.104541).

Data availability

Data will be made available on request.

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